

ART, GEOMETRY, AND COGNITION

# Entering Las Meninas

How a Painting Changed the Way I Look at Art

Zhuang Chengbo · Final Project · Summer 2026

UC Berkeley



# Why Las Meninas?

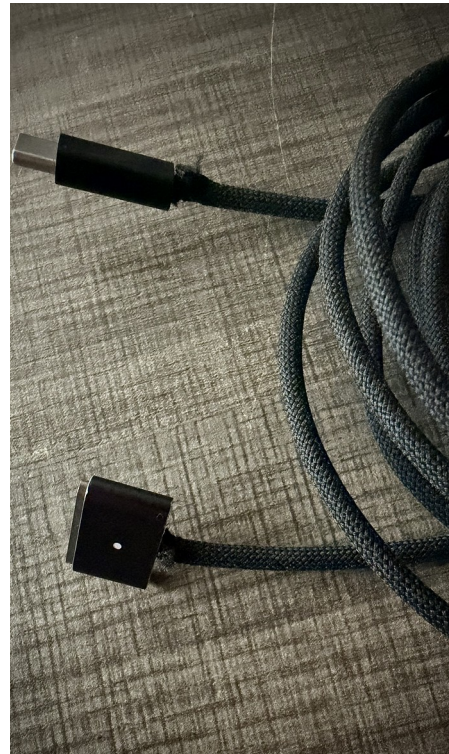
**“It did not let me remain outside the scene.”**

- The painter and the large canvas block a complete view.
- The mirror and rear doorway open the room beyond itself.
- Outward gazes make the viewer part of the composition.

**LOOKING BECAME PARTICIPATION.**



# Tether: a ready-made self-portrait



An ordinary object

- Came with my first university Mac
- Both ends frayed within one year
- Its wear records where I stood: before a screen

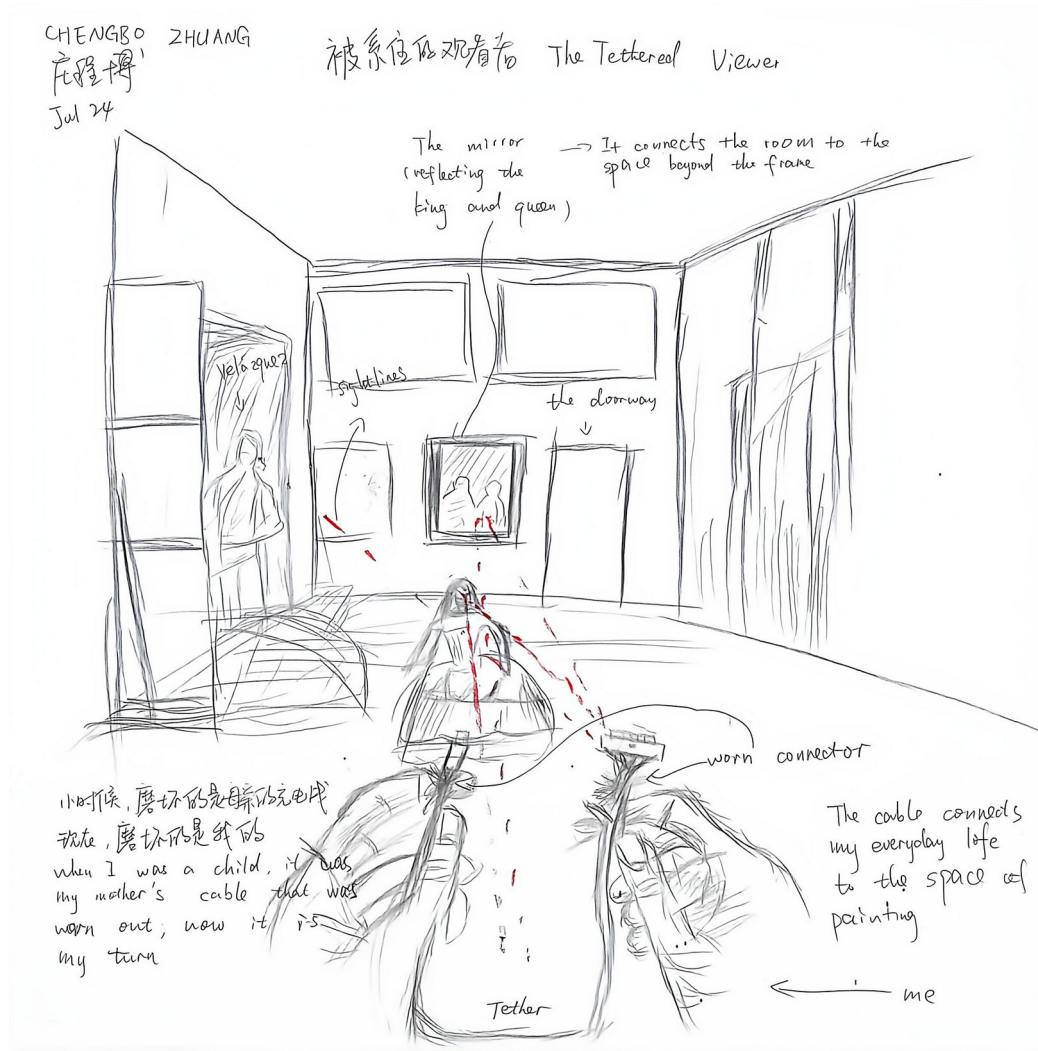
SELECTION + CONTEXT = MEANING



# The Tethered Viewer

- A room derived from Las Meninas
- My hands occupy the foreground
- The red tether crosses the picture's depth

childhood → university  
viewer → painting  
object → digital life

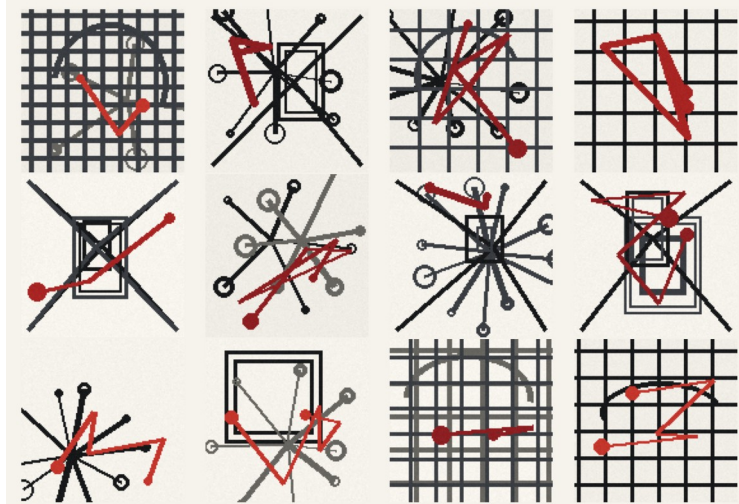


# A deliberately small neural experiment

GEOMETRIC TRAINING VOCABULARY

ENCODER → BOTTLENECK → DECODER

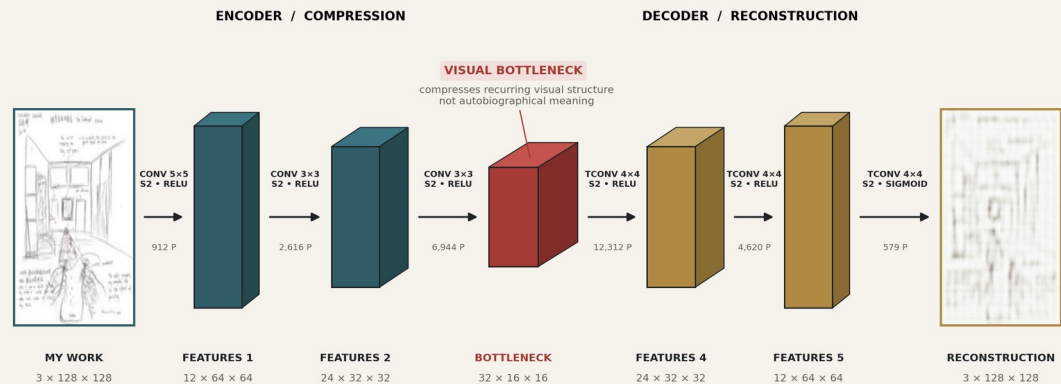
Synthetic Training Set — frames, perspective, curves, and red connections



The network never trains on Las Meninas or on my drawing.

## WHAT SURVIVES THE BOTTLENECK?

A 27,983-parameter geometric convolutional autoencoder



ONE FORWARD PASS 49,152 input values → 8,192 bottleneck activations → 49,152 reconstructed values

No classifier • no semantic labels • no skip connections • trained only on synthetic geometry

800  
synthetic patterns

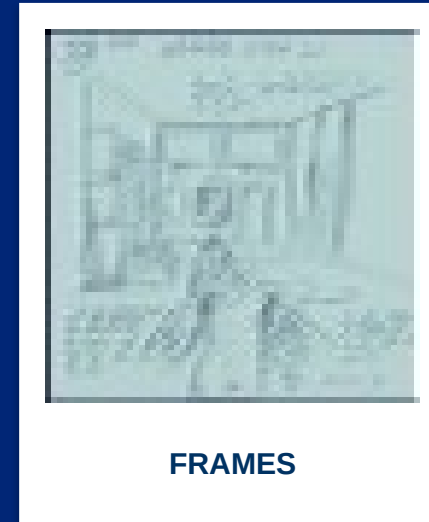
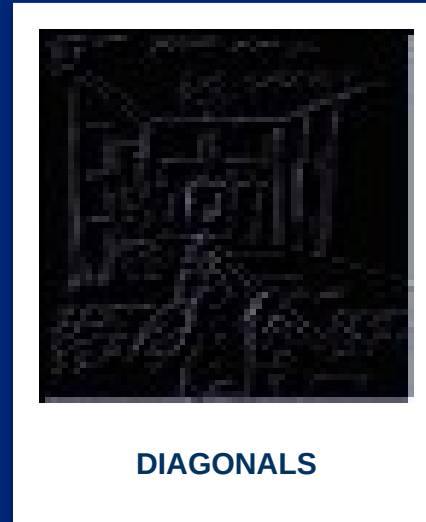
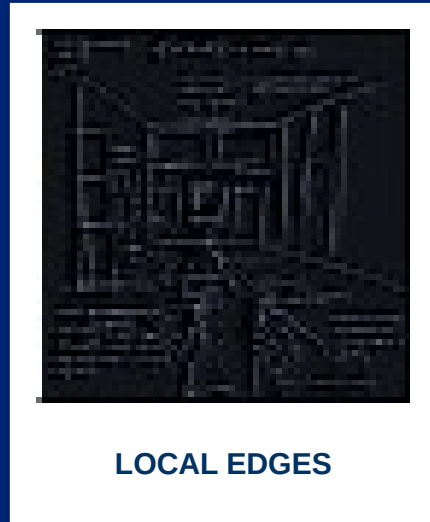
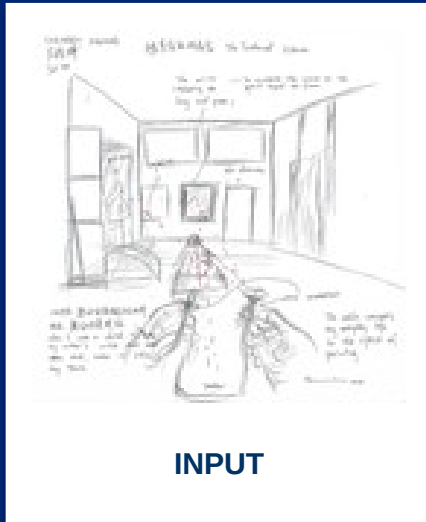
27,983  
parameters

128 × 128  
input

18 epochs  
MSE objective

# Early maps separate visual tendencies

12 × 64 × 64



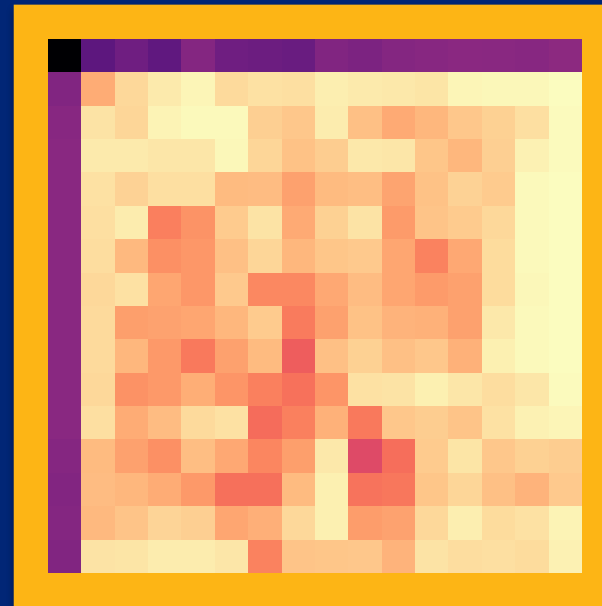
12 channels ≠ 12 meanings

The filters respond to structure, not to my story.

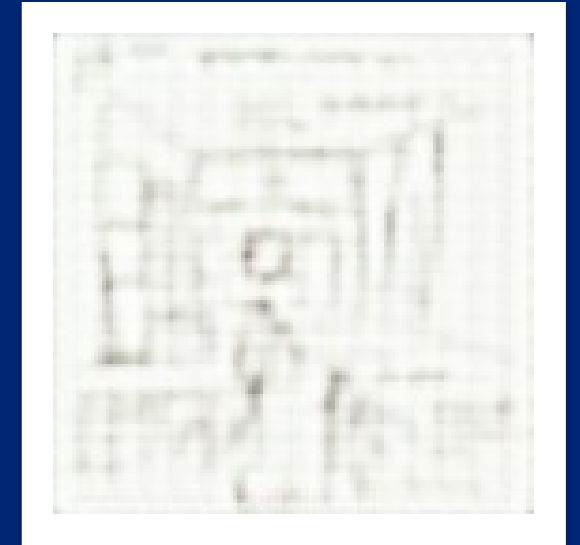
# What survives compression?



INPUT · 49,152 VALUES



BOTTLENECK ·  $32 \times 16 \times 16$   
8,192 VALUES



RECONSTRUCTION

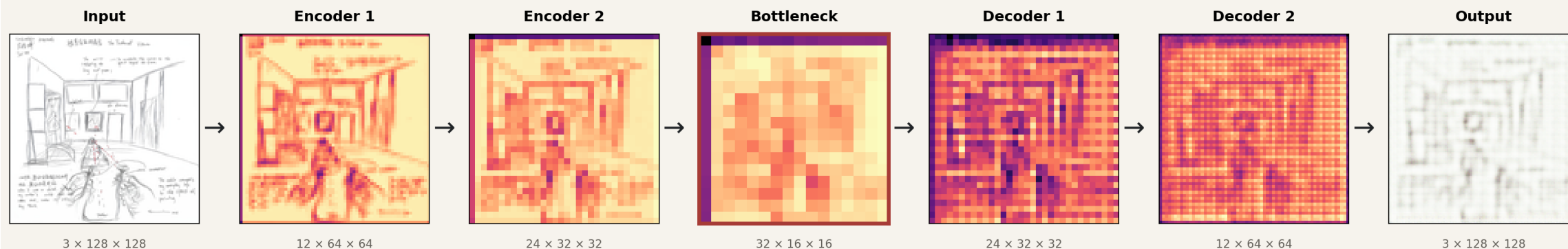
PRESERVED frames · openings · diagonals · approximate figures

DISCARDED handwriting · faces · exact red tether

# Decoder: expansion is not recovery

$128^2$  INPUT  $\rightarrow 64^2 \rightarrow 32^2 \rightarrow 16^2$  BOTTLENECK  $\rightarrow 32^2 \rightarrow 64^2 \rightarrow 128^2$  OUTPUT

The Tethered Viewer — one real forward pass through every scale



Heatmaps show the mean activation across channels and are normalized independently. They are not attention maps or semantic explanations.

## RECOVERED

Room frame · central rectangle  
approximate figures · major directions

## NOT RECOVERED

Handwriting · faces · fine detail  
exact shape of the red tether

MSE 0.003884  
MAE 0.048645

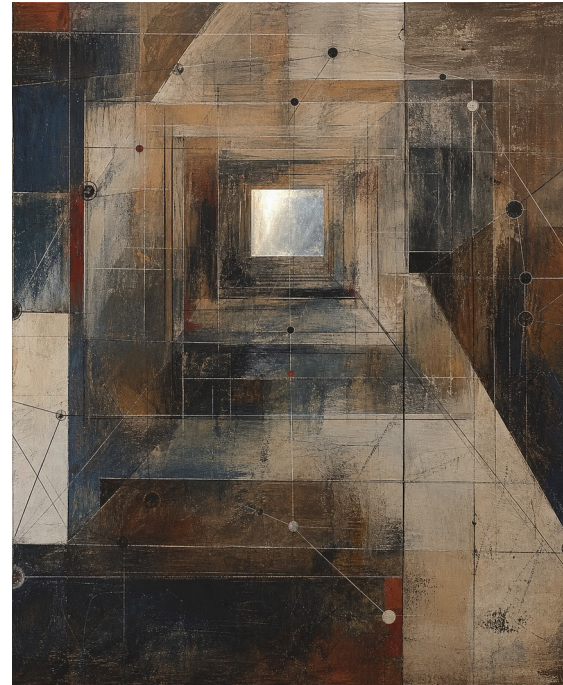


# Making the geometry of looking visible

CONTENT · LAS MENINAS



STYLE · COGNITIVE MAP



FINAL STYLE-TRANSFER COMPOSITION



figures + room → planes + nodes + convergence

# A four-second step inward

## HOW IT WAS MADE

- heuristic vertical depth field
- smooth sine/cosine camera loop
- depth-weighted pixel remapping

## WHY IT FEELS IMMERSIVE

Foreground drift + a stable doorway create parallax and layered depth.

**Every way of seeing has a bottleneck.**

4 s LOOP

